PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT AUTHORITY
KINONDONI MUNICIPAL COUNCIL
FORM FOUR DISTRICT PRE-MOCK EXAMINATION

CODE: 033/1 **BIOLOGY**

Time: 3 Hours **Year: March 2024**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and any **TWO (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. All writing should be in black or blue pen, except fr diagrams that must be drawn in pencil
5. Cellular phones and any unauthorized materials are not allowed in the examination room.
6. Write your examination number on every page of your answer booklet(s)

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SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items **(i) – (x),** choose the correct answer among the given alternatives and write its letter in the booklet provided.

(i) Mr. Chacha, A form One biology teacher at Mkalawa Secondary School set experiment to demonstrate the effect of fertilizer on plant, growth. He applied right amount of fertilizer on farm A and farm B left without any treatment. What does farm A represent.

A. Hypothesis
B. Control group
C. Experiment group
D. Dependent variable
E. Controlled variable

(ii) During prolonged fasting, in what sequence are the following organic compounds used up by body?

A. First fat next carbohydrates and lastly protein
B. First carbohydrate next protein and lastly fats
C. First proteins, next lipids and lastly carbohydrate
D. First carbohydrates, next fats and lastly proteins
E. First proteins, next carbohydrates and lastly fats.

(iii) .....is a structure that covers the larynx automatically during swallowing:

A. Trachea
B. Sphincter
C. Epiglottis
D. Tongue
E. Pharynx

(iv) Plants do not need specialized excretory system like animals because of the following reasons, except?

A. Have large vacuole which store wastes
B. They produce less poisonous substances
C. They have cell wall made up of cellulose
D. They can store the waste in organs that are destined to fall off or die off
E. Their rate of producing waste products is very small

(v) Juma has blood group O. This means Juma has:-

A. Antigen A and antibody b
B. No antigen but antibodies a and b
C. Antigen O, antibodies a and b
D. Antigen AB, but no antibody
E. Antigen B, antibody a

(vi) Which of the following is an example of a mutualistic relationship?

A. A caterpillar eating tomato plant
B. A flowering plant and its pollinators
C. A plasmodium inside the human body
D. A tick and a cow
E. A lion and antelope

(vii) Medical waste is described as .....

A. Biohazardous
B. Reusable materials
C. Biodegradable
D. Recyclable
E. Non degradable

(viii) Mtasha was passing in a forest and heard a voice of lion. Which of the following components occurred correctly in Mtasha's body;-

A. Receptor, effectors, stimuli, coordinator, response
B. Stimulus, receptor, coordinator, effector, response
C. Effector, stimulus, receptor, coordinator, response
D. Response, receptor, coordinator, effector, stimulus
E. Coordinator, response, stimulus, effector, receptor

(ix) Which is true about Kingdom?

A. A rank with most related organisms
B. A second rank of classification
C. A rank with an organism which can interbreed freely to produce a viable offspring
D. A rank with a few number of organisms
E. A rank with a most unrelated organism

(x) Which of the following methods dehydrates microbial cells by plasmolysis thereby killing them;-

A. Salting
B. Smoking
C. Refrigeration
D. Pasteurization
E. Canning

2. Match the number of the items in **LIST A** against the corresponding letter of item from **LIST B.**

LIST A		LIST B
(i) E	Defensive ability of body to fight infection which occur independent of human intervention	A. Artificial Immunity B. Natural Immunity C. Active Immunity D. Active artificial Immunity E. Passive artificial Immunity F. Active natural Immunity G. Passive natural Immunity
(ii) A	Vaccination of people against diseases such as polio, measles etc in which dead or alternated pathogens are injected into the body to stimulate antibodies production	
(iii) B	When mothers antibodies are passed to the baby through placenta or during breast feeding to offer protection	
(iv) C B	Resistance against infections that develop as the results of individual's exposure to an infection	
(v) D	When the body manufacture their its own immunological cells or antibodies	
(vi) F	Introduction to immunological cells or antibodies from resistance organism to an individual in order to protect him/her example tetanus vaccine	

SECTION B (54 Marks)
Answer all questions in this section.

3. (a) Briefly explain the process cell differentiation and state its significance.
   (b) Draw a diagram of an onion cell and label the following parts.
      (i) The full permeable membrane that encloses the inner parts of the cell
      (ii) The site of photosynthesis
      (iii) A large cellular organelle that contains hereditary materials.

4. (a) What would hypothetically likely to happen to an ecosystem, in case the following organisms are completely eliminated?
      (i) Green plants
      (ii) Bacteria
   (b) (i) Biological methods are preferred to chemical methods in the control of pests and parasites. Give three reasons.
      (ii) It is possible to contract amoebic dysentery as a result of close contact with infected person. Similar contact with a malaria victim can not result in infection. Explain.

5. (a) Explain the following phenomena:
      (i) Why the rate of breathing increases quickly during physical exercise?
      (ii) Why it is better to breathe through the nose than through the mouth?
   (b) Tabulate four differences between aerobic respiration and anaerobic respiration.

6. (a) Describe four major characteristics of the genetic material.
   (b) Outline five (5) differences between DNA and RNA.

7. (a) Explain three (3) conditions necessary for a viable seed to germinate.
   (b) Outline the changes that take place inside a seed before it germinate.

8. Explain six (6) causes of infertility in male.

SECTION C (30 Marks)
Answer any **Two (2)** questions from this section.

9. Different animals have different ways of regulating their body temperatures. Explain six (6) ways by behavioural means where poikilotherms can adjust their body temperature.

10. Some fungi are useful and others are detrimental to man. Explain this fact and give one example in each case. Give six points.

11. Describe how a finned fish is adapted for locomotion in water. Give six (6) points.